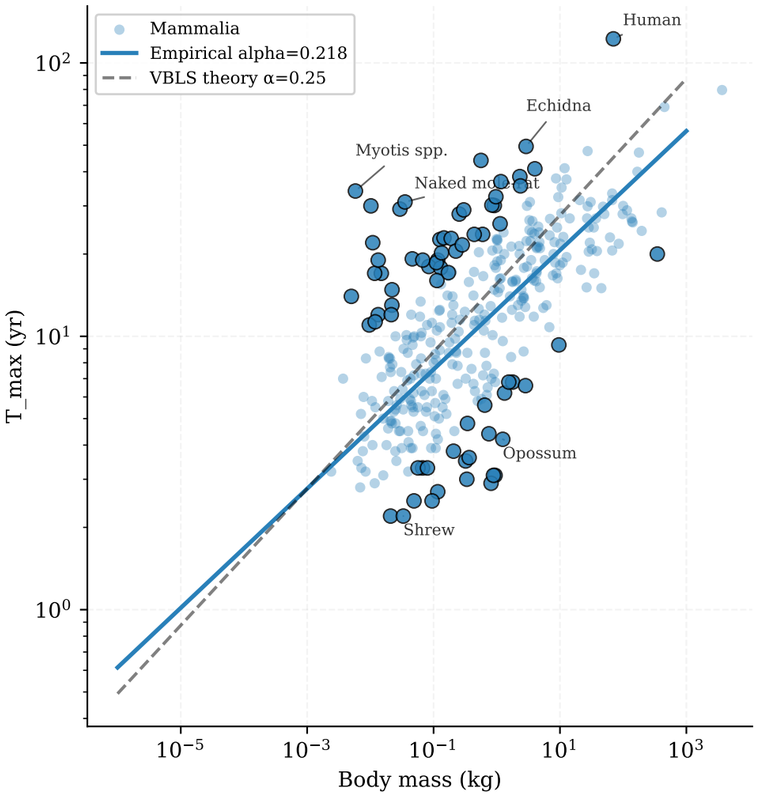
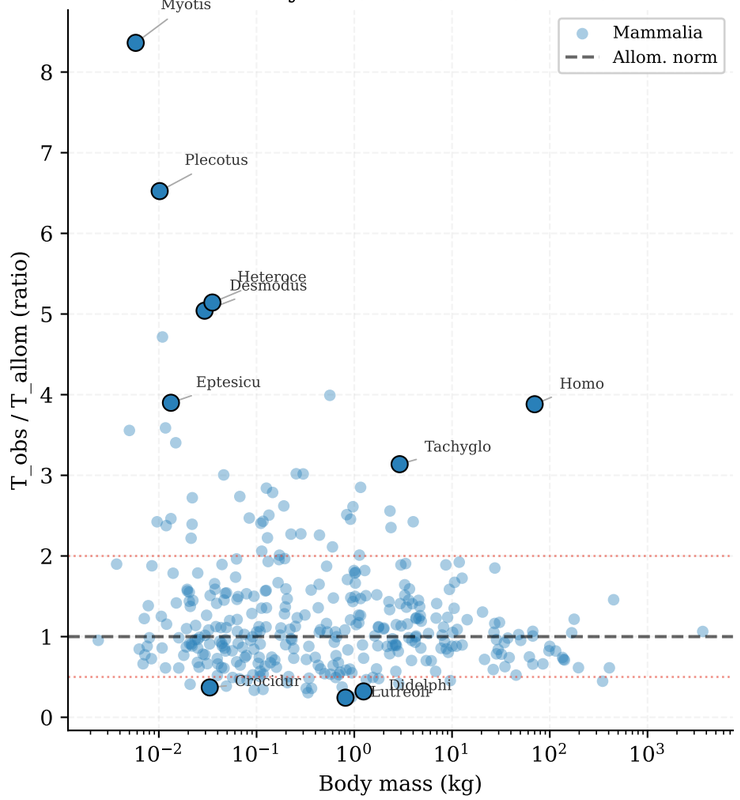
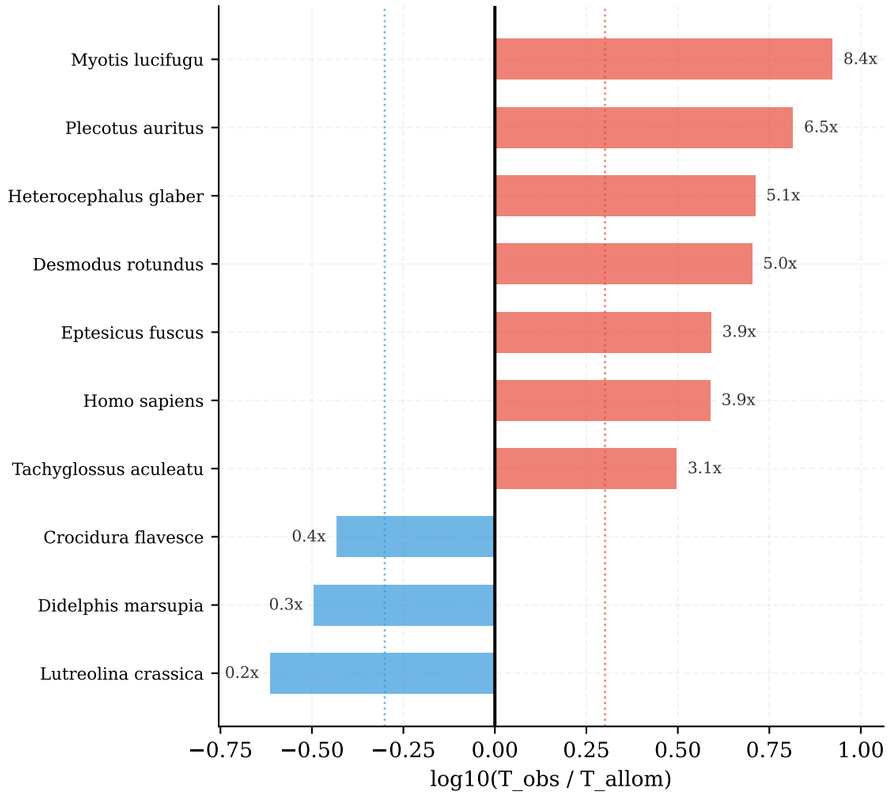
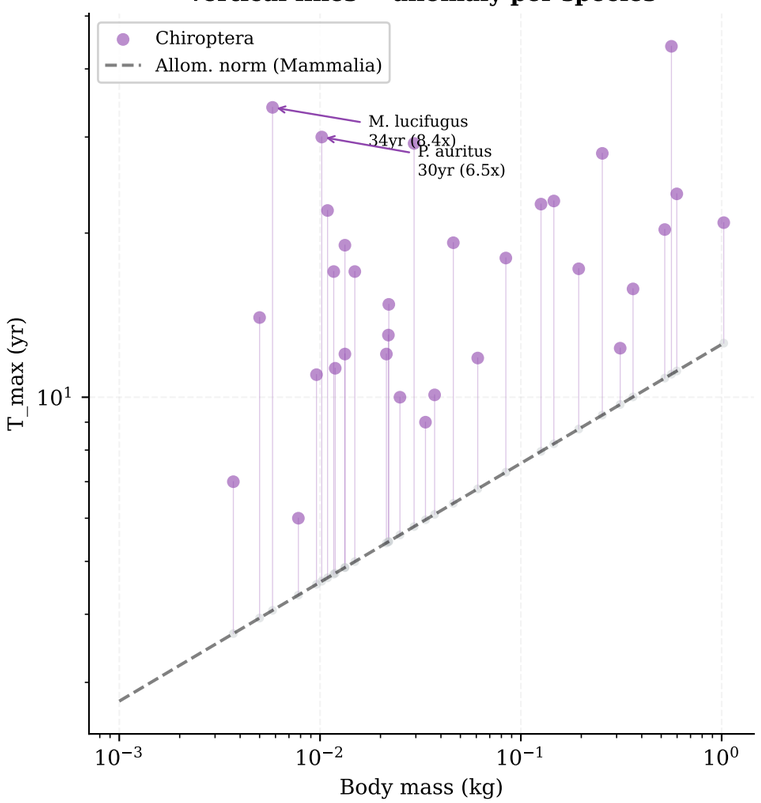






D. Anomaly typology in terms of VBLS
Each type corresponds to one parameter of system (S)

Type	Ratio	Parameter (S)	Biological mechanism (ex
High metabolism	<1	sigma0 up	High ROS, fast DNA wear (shrews, opossums)
Strong repair (+)	>2	rho up	Highly efficient DNA repair (naked mole-rat, Homo)
Low metabolism (	>2	sigma0_eff down	Antioxidants, slow turnover (tortoises, ectotherms)
Hibernation (+)	>2	nu0 down (seas.)	Effective < chronological lifespan (bats: Myotis, Plecotus)
Lowered body T (	>2	nu0 down (T)	van t Hoff: x2-3 per 10C (echidna, 30-32C)
r-strategist (-)	<0.5	rho down	Evolutionarily low repair invest. (opossums, mice)



F. Summary: each anomaly -> one VBLS parameter

```
VBLS interpretation of anomalies: key conclusions

^ ANOMALOUSLY LONG-LIVED:

Chiroptera (Myotis, x8.4):
  sigma0_eff down from flight — AMPK/FOXO3 activation
  of antioxidants reduces ROS production despite
  high metabolism. The flight paradox.

Heterocephalus glaber (x5.1):
  rho up — PARP1 x3, ribosomal fidelity up,
  hypoxic adaptation -> low ROS

Homo sapiens (x3.9):
  rho_eff up via medicine and social protection;
  biological rho not unique, but cultural
  'additional rho' is species-specific

Tachyglossus aculeatus (x3.1):
  nu0 down — body temperature 30-32C against 37C
  in placentals; van't Hoff slows degradation

v ANOMALOUSLY SHORT-LIVED:

Crocidura (x0.37):
  sigma0 up — metabolism 35 W/kg, heart
  1200 bpm -> maximal ROS

Didelphis (x0.32):
  rho down — r-strategist, evolutionarily optimized
  for reproduction, not longevity

-----------------------------------------
All anomalies -> one parameter of system (S).
Mammalia only: one class, one baseline.
```